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Ransomes Jacobsen Ltd, Textron Specialized Vehicles

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Robotics Engineer, Ipswich, Suffolk

Dear Hiring Manager,

The line about supporting the transition from prototype systems to field-ready capability is what made me read the rest of your listing. Making something survive outdoors, unattended, in weather, is the part of robotics I have spent most of my career on, and it is rarely the part that gets advertised.

I will be direct about where I fall short. You ask for five years in robotics or autonomy. I have just over three commercial, plus an MSc in Robotics from Bristol. Localisation and sensor fusion is genuinely my ground. Planning is not: my Nav2 work is integration and configuration, not algorithm authorship, and I would rather say so now than have it surface in an interview.

Most of my localisation experience comes from my MSc dissertation, a real-time SLAM system in C++17 and ROS2 on a Clearpath Husky that fused Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing. I calibrated the LiDAR extrinsics to the robot base and then spent an unglamorous stretch of weeks resolving tf2 transform timing across the point cloud pipeline, which is usually where localisation quietly goes wrong long before anyone blames the algorithm. I have kept at that work since, building a ROS2 Jazzy and Gazebo Harmonic stack to compare LiDAR-only against LiDAR and RGB fusion SLAM, with an EKF layer in robot_localization fusing wheel odometry and IMU. Benchmarking RTAB-Map, LIO-SAM, Cartographer and GLIM over the same data under one protocol taught me more about how each backend degrades than any paper has. Nav2 in simulation is how I check survey routes before committing a robot to them.

Real-time performance is the other half of what I do. At LivNSense I led four engineers on asynchronous C++17 perception pipelines on Intel edge hardware and took throughput from 5 to 15 FPS, mostly by overlapping preprocessing, inference and post-processing and following Intel VTune to the actual hotspots rather than the ones I assumed. The MSc stack ran on a Jetson AGX Xavier, inside the compute the platform had rather than the compute I wanted.

Field data is where I have put the most hours. At Esbaar I owned perception on BP drilling rigs running around the clock in desert conditions, led the on-site validation, and cut live pipeline crashes by 67% with health checks, safe restarts and a watchdog that recovered failed processes without anyone driving out to site. That fallback logic sat at the service level rather than the localisation level, but the habit transfers: decide what happens when a sensor drops out before it drops out.

RTK GNSS, coverage path planning and MPC I have not shipped, and I would need ramp-up on all three. I am UK based and happy to talk about relocating to Ipswich. I would welcome a conversation.

Yours sincerely,

Dexter Fernandes